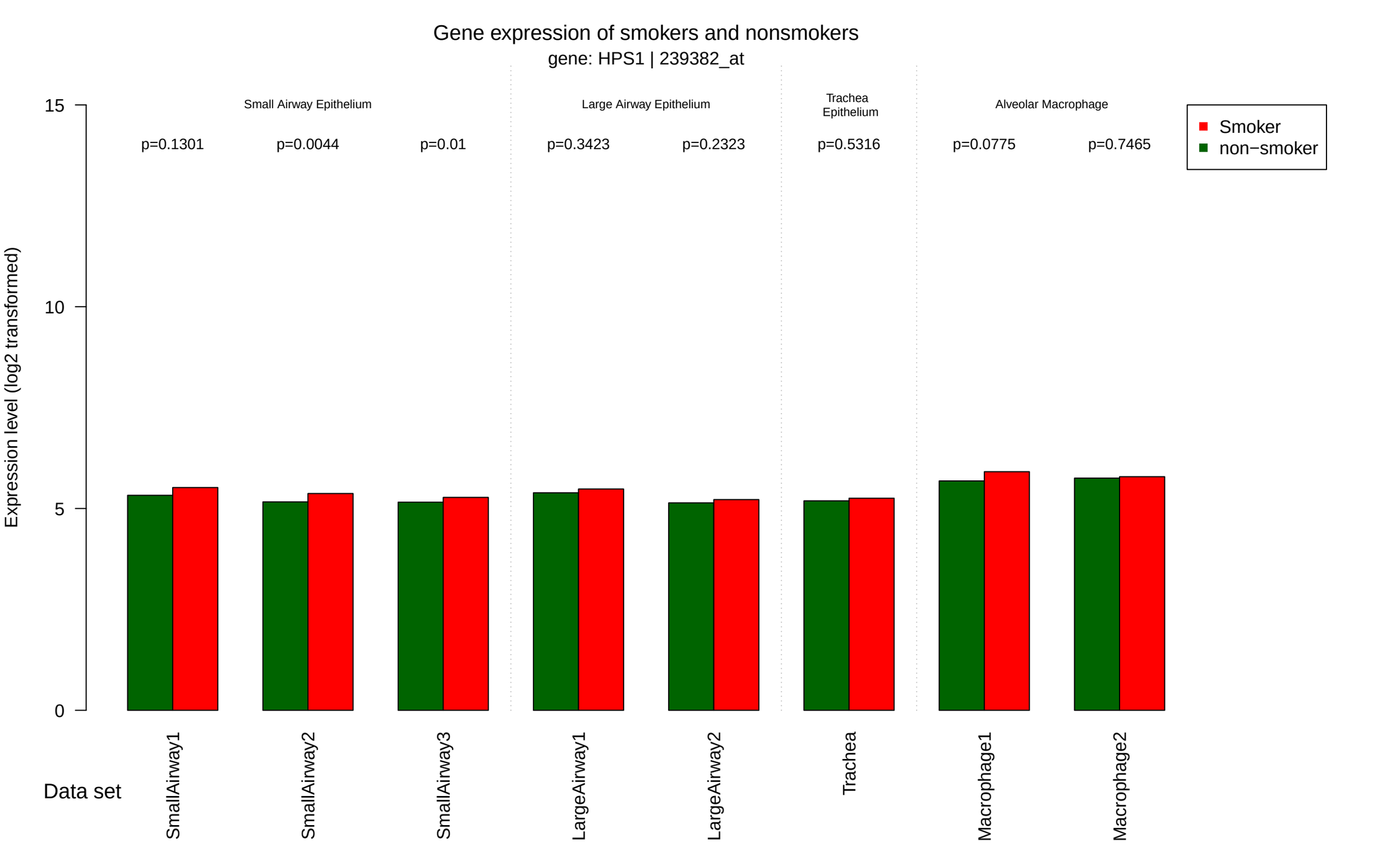
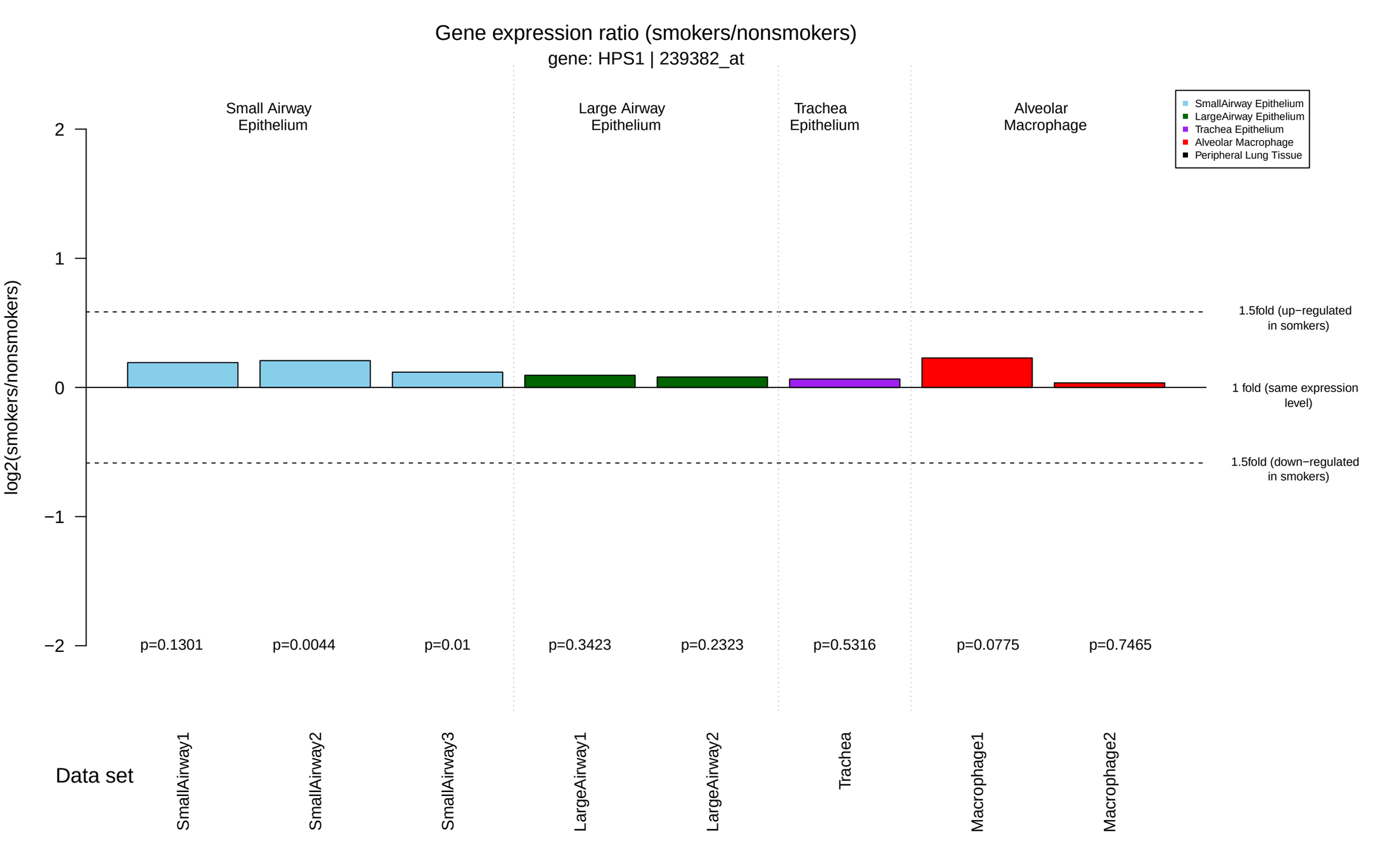
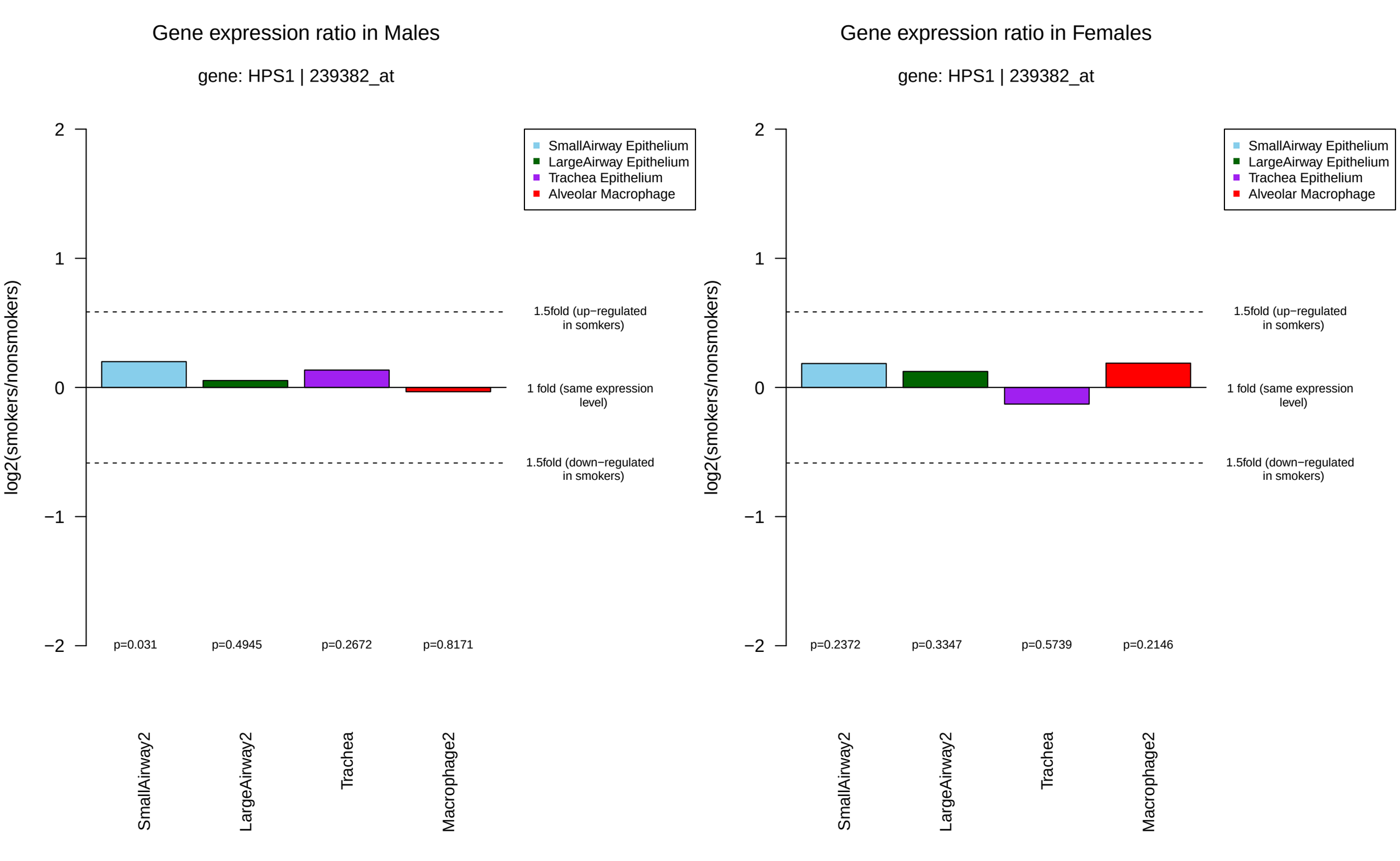
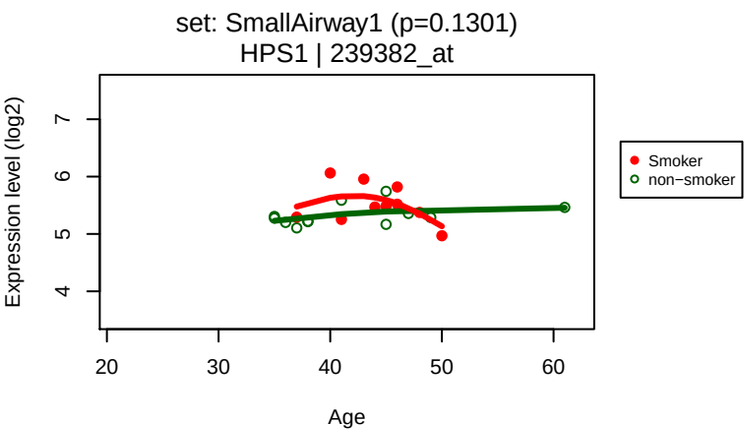
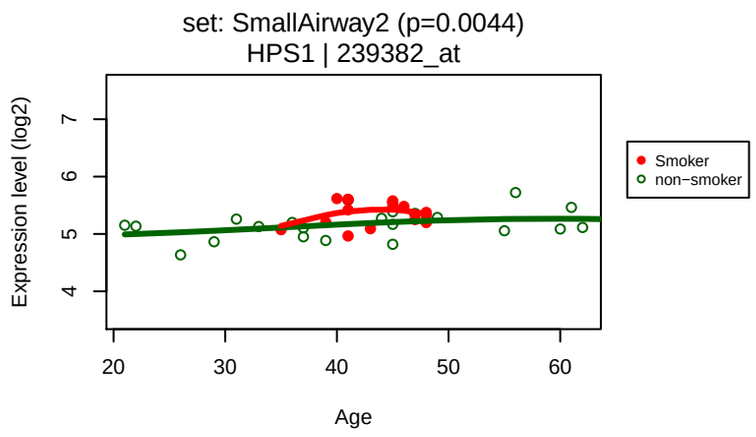
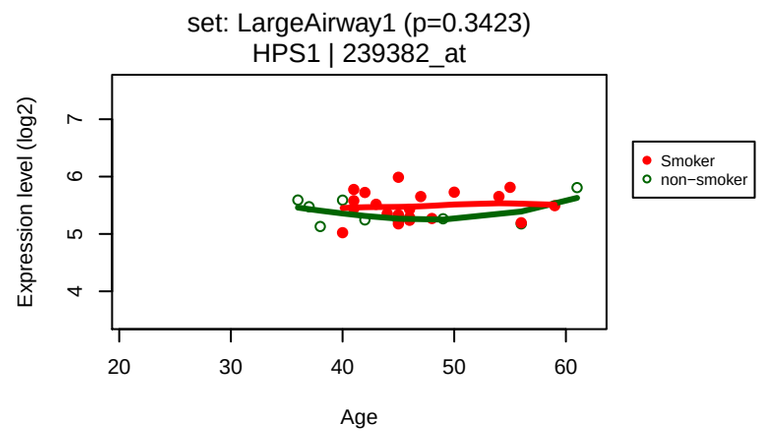

Gene expression of smokers and nonsmokers
gene: HPS1 | 239382_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1301
p=0.0044
p=0.01
p=0.3423
p=0.2323
p=0.5316
p=0.0775
p=0.7465
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

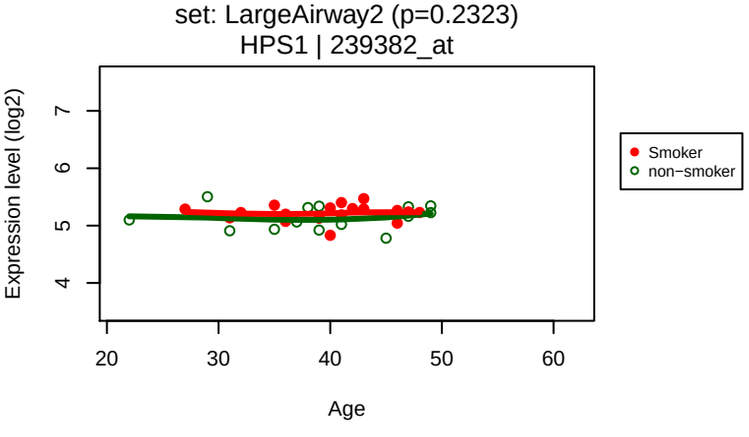

Gene expression ratio (smokers/nonsmokers)
gene: HPS1 | 239382_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1301
p=0.0044
p=0.01
p=0.3423
p=0.2323
p=0.5316
p=0.0775
p=0.7465
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

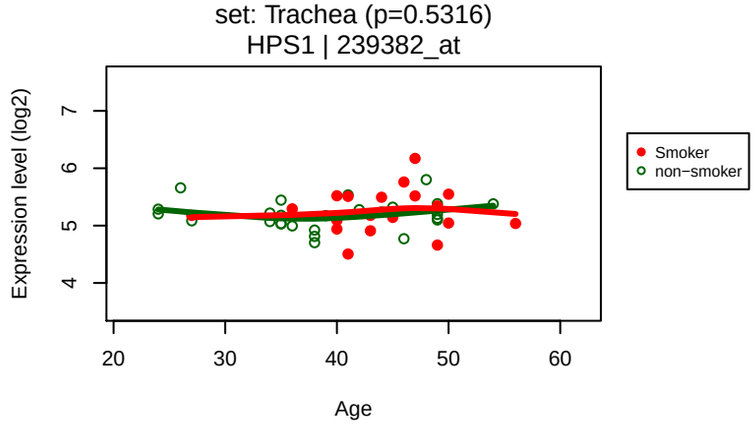

Gene expression ratio in Males
gene: HPS1 | 239382_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.031
p=0.4945
p=0.2672
p=0.8171
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPS1 | 239382_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2372
p=0.3347
p=0.5739
p=0.2146
SmallAirway2
LargeAirway2
Trachea
Macrophage2

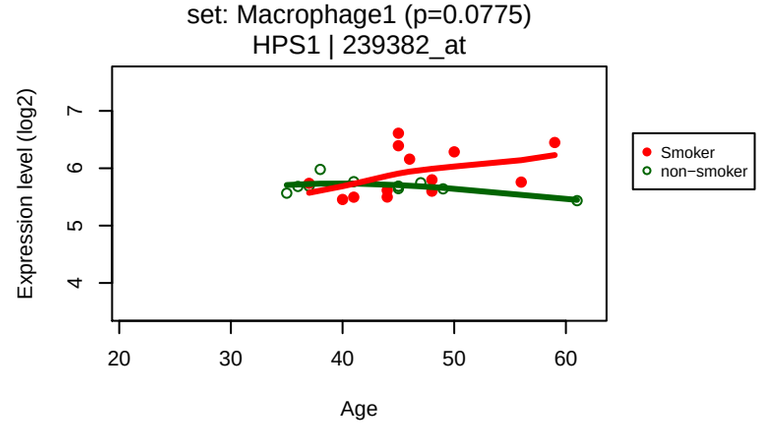

set: SmallAirway1 (p=0.1301)
HPS1 | 239382_at
Expression level (log2)
Age
Smoker
non−smoker

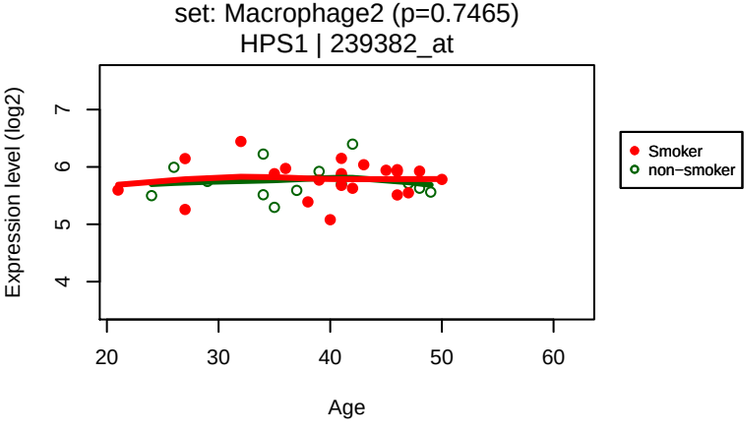

set: SmallAirway2 (p=0.0044)
HPS1 | 239382_at
Expression level (log2)
Age
Smoker
non−smoker

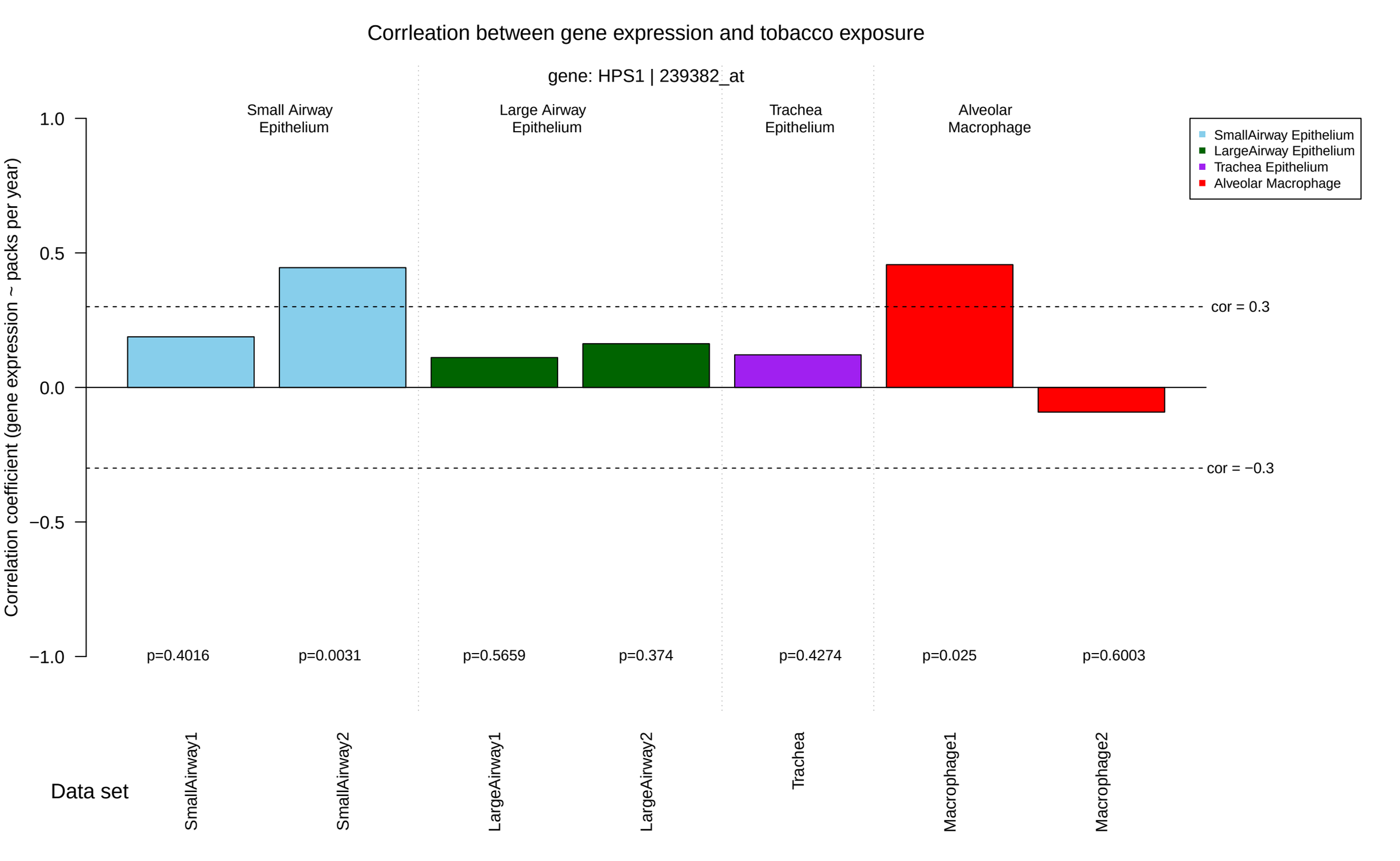

set: LargeAirway1 (p=0.3423)
HPS1 | 239382_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2323)
HPS1 | 239382_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5316)
HPS1 | 239382_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0775)
HPS1 | 239382_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7465)
HPS1 | 239382_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPS1 | 239382_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4016
p=0.0031
p=0.5659
p=0.374
p=0.4274
p=0.025
p=0.6003
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2